

# Two kinds of AI in research math

A Millennium Prize claim, and an afternoon with a coding assistant

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# Why this talk

- 8 September: OpenAI says its AI settled a 90-year-old, \$1M question about how fluids move.
- The night before: two mathematicians, working with Claude and Codex, go public with a year of closely related work.
- Two weeks ago: I rebuilt the simulation methods from my 2016 dissertation with Claude Code, in a day.

## Thesis

"AI in research math" currently means two very different things. Both are worth seeing up close.

**Part 1 (15 min).** The Navier-Stokes claim: what was claimed, how it was produced, who else was there, what is still open.

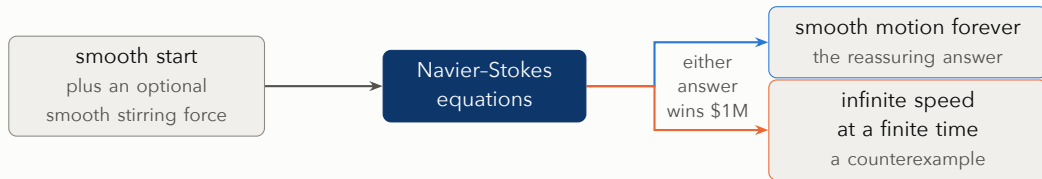
**Part 2 (15 min).** Wave simulations from my dissertation, re-derived with an assistant. Two short videos.

## **Part 1: the Navier-Stokes claim**

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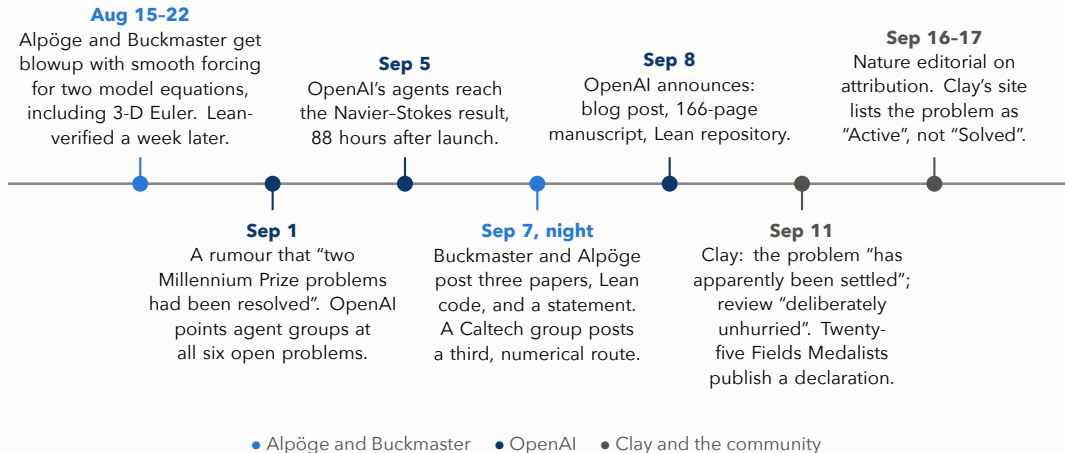
# Fluids, Newton, and a question from 1934

- The **Navier-Stokes equations** are Newton's  $F = ma$  for a fluid: water, air, blood. They sit inside aircraft design, weather forecasting, and ocean models.
- Leray's question (1934): from a smooth start, do the equations always predict smooth motion, or can they predict **infinite speed in finite time** ("blowup")? Real fluids cannot; blowup would mean the model leaves its own range of validity.
- Clay Mathematics Institute, 2000: \$1M for an answer either way. The fine print: a counterexample may use a smooth, well-behaved stirring force. The picture most people carry, "can turbulence by itself blow up?", is the *unforced* question.



Source: C. Fefferman, *Existence and smoothness of the Navier-Stokes equation*, Clay Mathematics Institute, p. 2.

# What happened, in two weeks

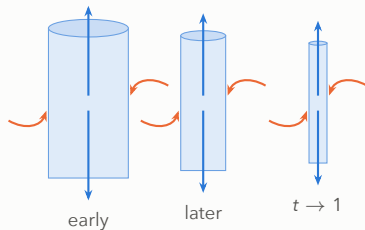


Sources: OpenAI's post; Buckmaster's statement; claymath.org; mathandai.org; nature.com. Every date is traced in docs/navier-stokes-notes.md.

# The claim, and the picture behind it

## Abstract, verbatim

"For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier-Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy."



After Fig. 1 of the manuscript. Orange:  
inward spiral; blue: outflow along the axis.

- Starts **at rest**; a smooth stirring force drives it. Energy stays bounded, yet the top speed reaches infinity at  $t = 1$ .
- A spinning column gets thinner, longer, faster ("like spaghetti"). Pulling mass inward speeds the spin; viscosity loses.
- The force holding the bare vortex together would itself blow up; fine **ripples** are added whose own motion cancels the bad part.
- **What it is not.** Nobody has shown that turbulence on its own blows up. The unforced question is untouched.

## How it was produced, in OpenAI's own account

|                                     |                                      |                               |                                              |                                                                   |
|-------------------------------------|--------------------------------------|-------------------------------|----------------------------------------------|-------------------------------------------------------------------|
| <b>~10,000</b><br>concurrent agents | <b>88 h</b><br>from launch to result | <b>130 B</b><br>output tokens | <b>17 h</b><br>more, to formalize in<br>Lean | <b>millions</b><br>of dollars, per Mark<br>Chen to <i>Science</i> |
|-------------------------------------|--------------------------------------|-------------------------------|----------------------------------------------|-------------------------------------------------------------------|

- An internal model “significantly more capable than GPT-6 Astra”, in training since 28 August.
- Groups of **coordinating agents** with tools (a cached copy of the internet, code execution), each group given one of the four official variants of the problem.
- Warm-up: the unforced Euler question fell first, to “nearly 100 agents” in about 50 hours. Everything was then pointed at Navier-Stokes, seeded with that solution.
- OpenAI will not claim the prize.

Source: OpenAI's post (8 Sep, updated 10 Sep); Science, 8 Sep. Nature and Science put the warm-up at 1,000 agents; the post says “nearly 100”. Check press numbers against the primary source.

# The parallel story, and the dispute

- Tristan Buckmaster (NYU) and Levent Alpöge (Anthropic), with Claude and Codex, for most of a year, on Buckmaster's own research funds. Both they and OpenAI build on Córdoba and Martínez-Zoroa.
- Results 15 August, Lean-verified 22 August, released 7 September, earlier than planned: "The Euler writeup, in particular, can only be described as AI slop."

## Buckmaster's statement (7 Sep)

- Emailed OpenAI on 3 Sep; two calls on 6 Sep. Told the proof came with "very little human input", which "turned out not to be true".
- "I asked whether the model had been trained on, or had access to, our sessions in Codex ... I did not get an answer."
- Declined two proposals, one a paper by Buckmaster alone crediting an OpenAI model, with Alpöge removed.

## OpenAI's post (8 Sep, updated 10 Sep)

- The effort "began on September 1st after hearing a rumor"; they "did not see any of their work through any means until they released it publicly".
- 8 Sep: "we cannot rule out that de-identified data ... helped improve our models"; 10 Sep, after an investigation: "could not have influenced the system in any way, including through training".
- "We recognize the priority of their work on forced Euler."

Neither account is independently verified: claims and responses, not findings. Sources: [cims.nyu.edu/~tristanb](https://cims.nyu.edu/~tristanb); [openai.com](https://openai.com).

# What “machine-checked” does and does not mean

## Lean checks

- That every step follows from three standard axioms. No gaps, no “clearly”.
- Reproducibly: clone the repository, run the independent checker.

## Humans still check

- That the formal theorem *is* the Clay statement. OpenAI targeted an independent transcription of it (Google DeepMind’s). Still a judgment.
- Whether the construction teaches anything. Clay: publication, a two-year wait, then a committee.

### Quanta, 8 Sep

The “crucial bit of verification that must still be done by humans is to guarantee that the statement being shown to be true in Lean is logically equivalent to what mathematicians set out to prove.”

# Reactions, and questions to argue about

## Reactions

- **Clay Mathematics Institute** (11 Sep): “apparently been settled”; evaluation “deliberately unhurried”. Site status: *Active*.
- **25 Fields Medalists** (11 Sep): “solving problems is only a tool and proxy for achieving the primary goal of conceptual understanding and insight”.
- **Terence Tao**: open problems “are being mined as a non-renewable resource”.
- **Tristan Buckmaster**: “This is a Deep Blue-Kasparov moment.”
- **Sébastien Bubeck** (OpenAI): “the spectacular culmination of the arc we have seen over the last 12 months”.

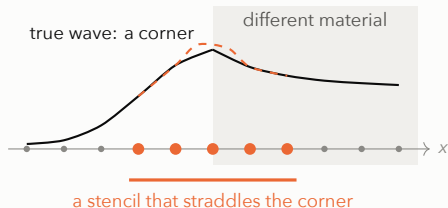
## Questions

1. **Letter vs. spirit.** The statement allows a smooth force, and this result uses one. Is the Millennium problem solved?
2. **Trust.** Would you put unpublished work into a frontier lab’s tools? What would you need to see first?
3. **Compute.** No university can run 10,000 agents for four days. What does that do to a field whose scarce resource was always ideas?

## **Part 2: the other kind**

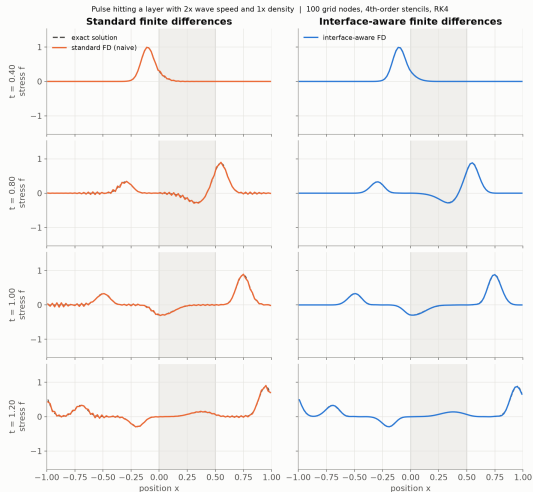
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# My corner of this world: waves through layered materials



- 2016 PhD, applied mathematics, CU Boulder: seismic imaging (sound through rock), ultrasound (sound through tissue).
- A simulation stores the wave at grid points and estimates slopes from a few neighbours: a **stencil**. One that straddles the corner assumes a smooth curve (dashed) and gets the slope wrong.
- The fix: rebuild the few stencils near the boundary from pieces that are allowed a corner there. Nothing else changes.
- The experiment: nine-year-old MATLAB, no licence. With Claude Code: re-derive from the papers, port to Python, verify against exact solutions. One day.

# 1-D: the ringing, and the fix

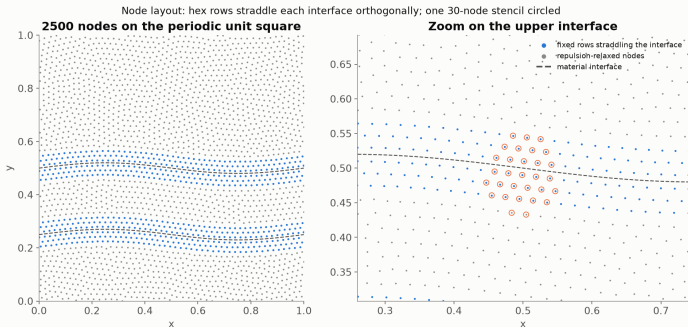


- A pulse hits a layer (shaded) where waves travel twice as fast. Part reflects, part transmits.
- Same 100 grid points, same time step, same code. Only the stencils touching the layer edges differ.
- **Standard**: wiggles behind every reflection; 10% error by  $t = 1$ .  
**Interface-aware**: on the exact solution; 3%, all of it from the coarse grid.

**Play clip 1** (10 s)

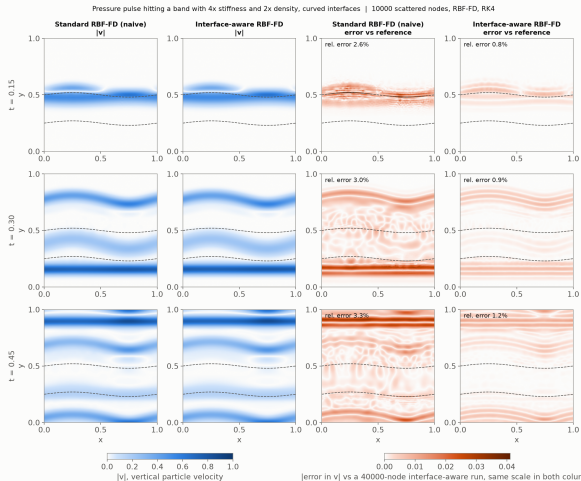
wave1d\_naive\_vs\_aware\_coarse.mp4

## 2-D: scattered points and curved boundaries



- No grid. Points are spread by a repulsion process, with fixed rows straddling each boundary (blue). That layout was the key to stability: years to find, still unproven. (2,500 shown; the runs use 10,000.)
- Each point's stencil is its 30 nearest neighbours, weighted with radial basis functions (RBF-FD). Five fields: two velocities, three stresses.

## 2-D: same points, only the stencils near the boundaries differ



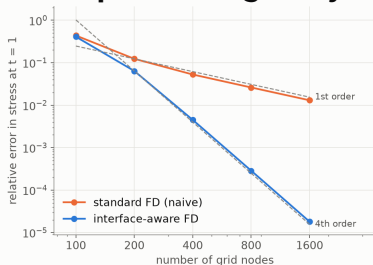
- A pressure pulse travels down through a curved band of stiffer, denser material. 10,000 points.
- Left pair: the wave. Right pair: error against a 40,000-point reference run, one colour scale.
- At  $t = 0.3$ : **standard** 3.0% error, **interface-aware** 0.9%. The standard error is born at the boundary and travels with the wave.

**Play clip 3 (8 s)**

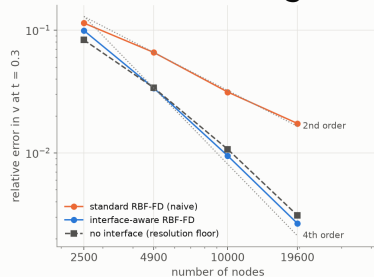
wave2d\_naive\_vs\_aware\_curved.mp4

# How fast the error shrinks

## 1-D: a pulse through a layer



## 2-D: elastic waves through a band



- **Standard**: double the points, halve the error (1-D) or quarter it (2-D). The corner caps the accuracy.
- **Interface-aware**: double the points, divide the error by 16. In 2-D it sits on the dashed no-boundary line: nothing left to fix.
- Same story for a thinner layer and for curved boundaries.

Python port of dissertation Figs. 2-8 and 3-5, error against exact solutions; thin-layer and curved cases in the repository.

# What the collaboration looked like

1

day (17 September)

9

pull requests

106

tests, all green

4

real bugs caught  
before the demo

2

methods, 1-D and  
2-D

## The assistant

- Every routine tested against an exact solution or a convergence rate, not “it runs”.
- Independent reference solutions built before trusting a solver.
- Review and cross-checks caught a sign error, two latent bugs in the exact solver that only the 2-D problem exposed, and a driver crash.

## Still human

- Where to put the points (the straddling rows). Plain method first, clever method second.
- Which basis functions, and the narrow window for the stabilising parameter.
- Steering, judging what matters, designing the verification. The typing, the bookkeeping, and the port were not.

## Two kinds of AI in research math

|                    | OpenAI, Navier-Stokes     | Alpöge and Buckmaster        | Me, with Claude Code          |
|--------------------|---------------------------|------------------------------|-------------------------------|
| Who steered        | a lab team, via prompts   | two mathematicians           | one engineer                  |
| Scale              | ~10,000 agents, 88 h      | about a year, several models | one assistant, one day        |
| Verification       | Lean; Clay review pending | Lean; papers being rewritten | tests against exact solutions |
| Who understands it | contested                 | "AI slop", being digested    | fully; it is my method        |
| Cost               | millions of dollars       | own research funds           | a subscription                |

- The capability is real at both ends of this table.
- The difference is who steered, who can check it, and who is left understanding the result.

# For the curious

## Gentle starts

- 3Blue1Brown, *But what is a partial differential equation?* (YouTube, 17 min)
- Khan Academy: [Differential equations](#)
- Wikipedia: [Navier-Stokes existence and smoothness](#); [Finite difference method](#); [Lean](#) (proof assistant)

## Primary sources for Part 1

- OpenAI: [the announcement and manuscript](#); [the Lean repository](#)
- Clay Mathematics Institute: [the problem page and official statement](#)
- [Buckmaster's statement](#) and papers ([cims.nyu.edu/~tristanb](https://cims.nyu.edu/~tristanb))
- [Tao's blog post, 7 Sep 2026](#); [the Fields Medalists' declaration](#)

## Part 2

- Code, tests, figures, the equations, and these notes:  
[github.com/bradleypmartin/20260930-zd-ai-pdes-demo](https://github.com/bradleypmartin/20260930-zd-ai-pdes-demo) (MIT)